Practice Set 2

1. Write a program to print the following pattern. Take a number of lines as input. For example, for input 5 the output is shown below:

```
    1
   1 1
  1 1 1
 1 1 1 1
1 1 1 1 1
```

2. Write a program to print the factorial of a number which is taken as input. Remember, the input can be anything between 1-100.
3. Given a number, check whether it is a strong number or not. A strong number is a special number, such that the sum of the factorials of all digits present in the number should be equal to the number itself.
4. Write a program to check whether a 3-digit number follows the given property or not. If the number follows it, print "Yes", otherwise print "No".
   Property: the sum of cubes of each digit is less than the number
   Example: 111 follows the given property as (1*1*1 + 1*1*1 + 1*1*1) < 111

5. 11. Check if the input phone number p is valid or not. A phone number is valid if,
   a. It contains 10 digits
   b. It may contain 11 digits, but in that case the first digit should be 0
   c. The starting digit should be either 9, 8 or 7

   Take p as input and output 1 if the phone number is valid else output 0.
6. Write a C program to check if a given number is an Armstrong number.

   Example: 153 → $1^3 + 5^3 + 3^3 = 153$

7. Write a C program to print all twin prime pairs between 1 and 100.

   Twin primes are pairs of prime numbers that differ by 2.

   Example: (3,5), (5,7), (11,13), (17,19)...

8. Write a C program that repeatedly displays a menu and performs the selected operation using a switch statement.
   Menu:
   1. Find GCD of two numbers
   2. Find LCM of two numbers
   3. Check if a number is Prime
   4. Exit

9. Write a C program to print the following pattern for a given number of rows n.

Example: For n = 5, the output should be:

```
    *
   * *
  *   *
 *     *
*********
```

10. Write a C program to print the following pattern for a given number of rows n.

Example: For n = 5, the output should be:

```
*****
 ****
  ***
   **
    *
   **
  ***
 ****
*****
```